

A decorative graphic on the left side of the slide consisting of two overlapping parallelograms. The front one is blue and the back one is light green. They are positioned diagonally, with the blue one partially covering the green one.

WATER QUALITY CLASSIFICATION USING LOGISTIC REGRESSION AND DECISION TREES

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Abstract

1. Research Objective

Goal: To evaluate and compare two supervised machine learning models—Logistic Regression and Decision Trees—for automated water potability classification.

Context: Address rising contamination risks with fast, automated, and robust water quality testing to safeguard public health.

2. Methodology & Features

Dataset: Utilized key physicochemical water parameters, including pH levels, turbidity, hardness, solids, and chemical contaminants.

Model Roles:

Logistic Regression: Used as an efficient, interpretable baseline for linear relationships.

Decision Trees: Deployed to capture non-linear patterns, complex feature interactions, and threshold-based decision rules.



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3. Key Findings & Performance

Evaluation Metrics: Accuracy, precision, recall, and F1-score.

Outcome: While Logistic Regression provided computational speed and probabilistic clarity, the Decision Tree model achieved superior overall performance due to its ability to handle intricate, non-linear environmental data.

4. Real-World Applications

Deployment: Confirms the viability of machine learning for real-time water quality monitoring.

Target Users: Offers municipal authorities and water treatment plants scalable, automated tools to streamline testing, prevent public health risks, and ensure regulatory compliance.



Introduction

1. The Problem & Significance

Vital Resource: Potable water is essential for life, ecosystems, and socioeconomic development, yet securing safe drinking water remains a major global challenge.

Pollution Threats: Industrialization, agricultural runoff, urbanization, and aging infrastructure introduce harmful pathogens, heavy metals, and chemical pollutants.

Health Risks: Contaminated water causes severe waterborne diseases and toxicological effects, making continuous monitoring essential.

2. Limitations of Conventional Testing

Traditional Approach: Relies on manual sampling followed by laboratory-based physicochemical and microbiological testing.

Drawbacks: Highly labor-intensive, costly, and slow, often causing critical delays in detecting contamination events.



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3. The AI & Machine Learning Approach

Modern Solution: Machine Learning (ML) provides rapid, automated, and scalable alternatives for predictive modeling and potability classification.

Algorithms Investigated:

Logistic Regression: Serves as a straightforward, interpretable baseline to identify linear relationships between water parameters and safety thresholds.

Decision Trees: A flexible, non-parametric model capable of capturing complex, hierarchical interactions and critical thresholds among parameters (such as pH, turbidity, and chemical concentrations).

4. Study Objective & Impact

Goal: Compare the predictive performance of both models to determine the most effective computational framework for real-time water quality assessment.

Practical Benefit: Enables water treatment facilities and regulatory bodies to detect contamination swiftly, allocate resources efficiently, and safeguard public health.



software/hardware requirements

- Operating System: Windows 10/11, Ubuntu 20.04+, or macOS (64-bit)
- Language & IDEs: Python 3.8+ using Jupyter Notebook, JupyterLab, or VS Code
- Key Libraries: Pandas, NumPy, Scikit-Learn, Matplotlib, and Seaborn.
- Hardware: Dual-core CPU (quad-core recommended), 4–8 GB RAM, and 2 GB storage.